

Digital Logic Design

Digital Systems and Binary Numbers



Binary-Coded Decimal (BCD)

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- BCD Code
 - A number with k decimal digits will require 4k bits in BCD.
 - **Decimal 396** is represented **in BCD** with 12bits as **0011 1001 0110**, with each group of 4 bits representing one decimal digit.
 - A decimal number in BCD is the same as its equivalent binary number only when the **number is between 0 and 9**.
 - The binary combinations **1010 through 1111** are not used and have no meaning in BCD.

Table 1.4
Binary-Coded Decimal (BCD)

Decimal Symbol	BCD Digit
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001

Binary-Coded Decimal (BCD)

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- Example:

- Consider decimal 185 and its corresponding value in BCD and binary:



$$(185)_{10} = (0001\ 1000\ 0101)_{\text{BCD}} = (10111001)_2$$

- BCD addition

4	0100	4	0100	8	1000
<u>+5</u>	<u>+0101</u>	<u>+8</u>	<u>+1000</u>	<u>+9</u>	<u>+1001</u>
9	1001	12	1100	17	10001
			<u>+0110</u>		<u>+0110</u>
			10010		10111

Binary-Coded Decimal (BCD)

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□ Example:

- Consider the addition of $184 + 576 = 760$ in BCD:

BCD	1	1		
	0001	1000	0100	184
	<u>+ 0101</u>	<u>0111</u>	<u>0110</u>	+576
Binary sum	0111	10000	1010	
Add 6	<u> </u>	<u>0110</u>	<u>0110</u>	<u> </u>
BCD sum	0111	0110	0000	760

- Decimal Arithmetic: $(+375) + (-240) = +135$

Hint: using 10's complement of BCD

0	375
<u>+ 9</u>	<u>760</u>
0	135

Binary Codes

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Other Decimal Codes

Table 1.5

Four Different Binary Codes for the Decimal Digits

Decimal Digit	BCD 8421	2421	Excess-3	8, 4, -2, -1
0	0000	0000	0011	0000
1	0001	0001	0100	0111
2	0010	0010	0101	0110
3	0011	0011	0110	0101
4	0100	0100	0111	0100
5	0101	1011	1000	1011
6	0110	1100	1001	1010
7	0111	1101	1010	1001
8	1000	1110	1011	1000
9	1001	1111	1100	1111
Unused bit combi- nations	1010	0101	0000	0001
	1011	0110	0001	0010
	1100	0111	0010	0011
	1101	1000	1101	1100
	1110	1001	1110	1101
	1111	1010	1111	1110

Gray Codes

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□ Gray Code

- The advantage is that only bit in the code group changes in going from one number to the next.
 - Error detection.
 - Representation of analog data.
 - Low power design.

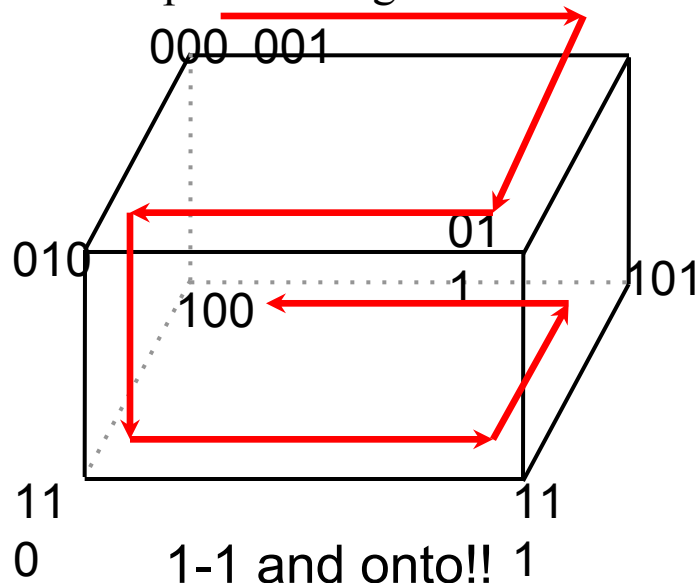
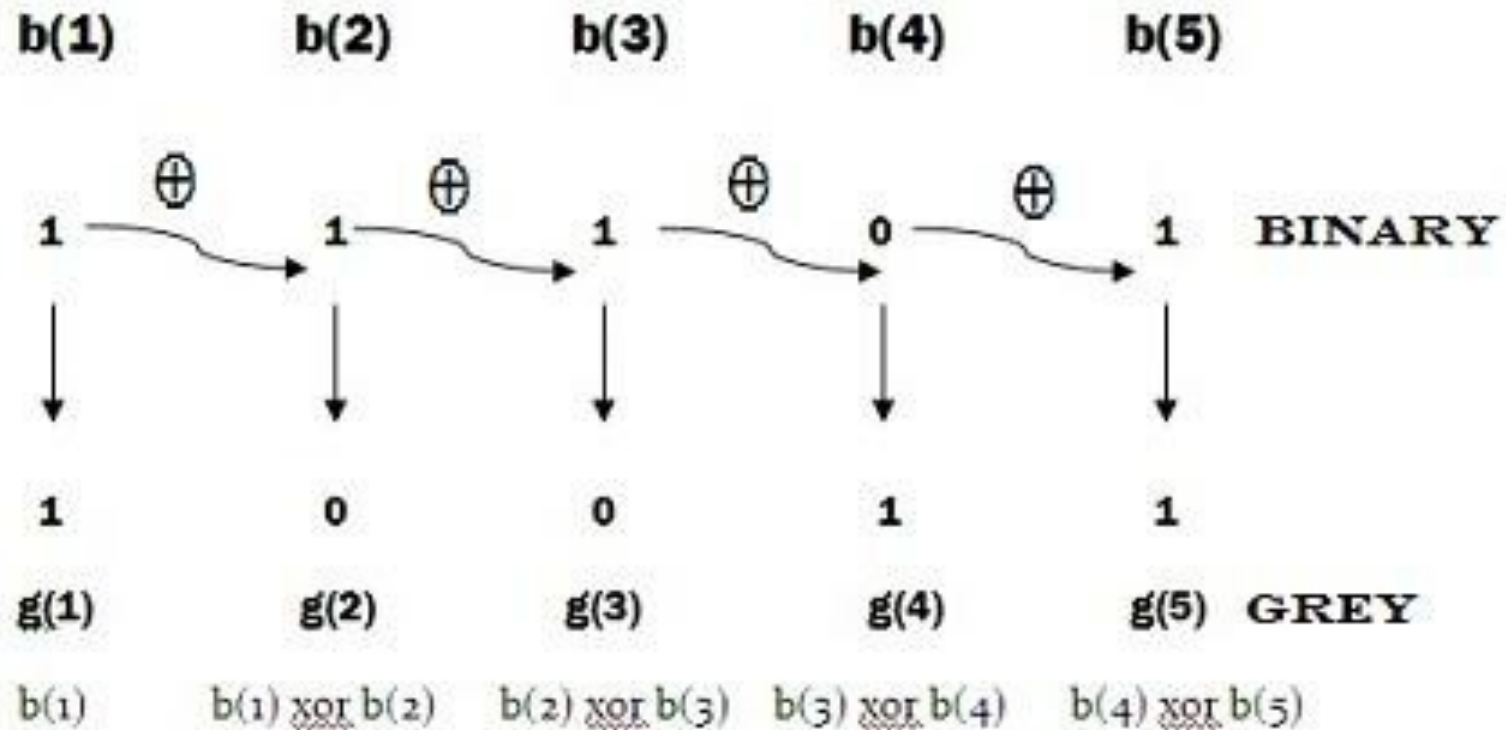


Table 1.6
Gray Code

Gray Code	Decimal Equivalent
0000	0
0001	1
0011	2
0010	3
0110	4
0111	5
0101	6
0100	7
1100	8
1101	9
1111	10
1110	11
1010	12
1011	13
1001	14
1000	15

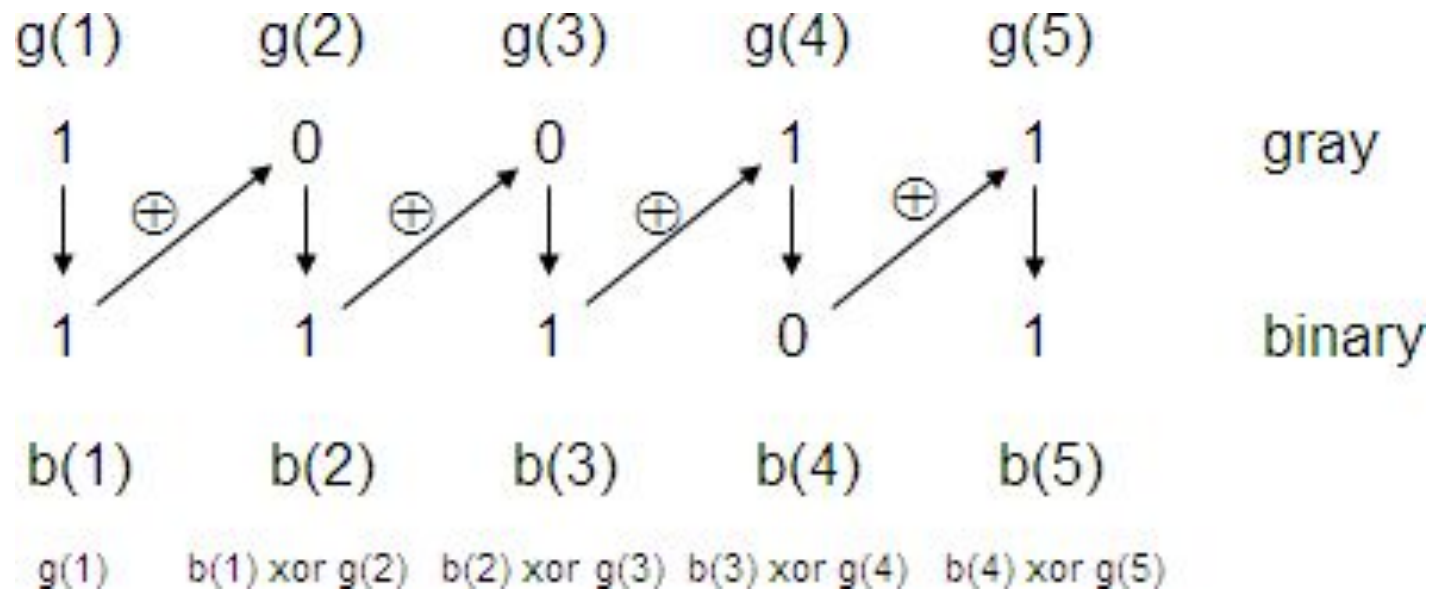
Gray Codes

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Gray Codes

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Binary Logic

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- Definition of Binary Logic
 - **Binary logic** consists of *binary variables* & a *set of logical operations*.
 - The *variables* are designated by letters of the alphabet, such as A , B , C , x , y , z , etc, with each variable having two distinct possible values: 1 and 0,
 - Three basic *logical operations*: **AND**, **OR**, and **NOT**.



Binary Logic

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1. AND: This operation is represented by a dot or by the absence of an operator. For example, $x \cdot y = z$ or $xy = z$ is read “ x AND y is equal to z ,” The logical operation AND is interpreted to mean that $z = 1$ if only $x = 1$ and $y = 1$; otherwise $z = 0$. (Remember that x , y , and z are binary variables and can be equal either to 1 or 0, and nothing else.)
2. OR: This operation is represented by a plus sign. For example, $x + y = z$ is read “ x OR y is equal to z ,” meaning that $z = 1$ if $x = 1$ or $y = 1$ or if both $x = 1$ and $y = 1$. If both $x = 0$ and $y = 0$, then $z = 0$.
3. NOT: This operation is represented by a prime (sometimes by an overbar). For example, $x' = z$ (or $\bar{x} = z$) is read “not x is equal to z ,” meaning that z is what x is not. In other words, if $x = 1$, then $z = 0$, but if $x = 0$, then $z = 1$, The NOT operation is also referred to as the complement operation, since it changes a 1 to 0 and a 0 to 1.



Binary Logic

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- Truth Tables, Boolean Expressions, and Logic Gates

AND

x	y	z
0	0	0
0	1	0
1	0	0
1	1	1

$$z = x \cdot y = xy$$



OR

x	y	z
0	0	0
0	1	1
1	0	1
1	1	1

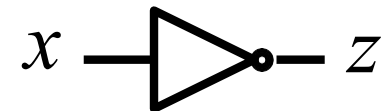
$$z = x + y$$



NOT

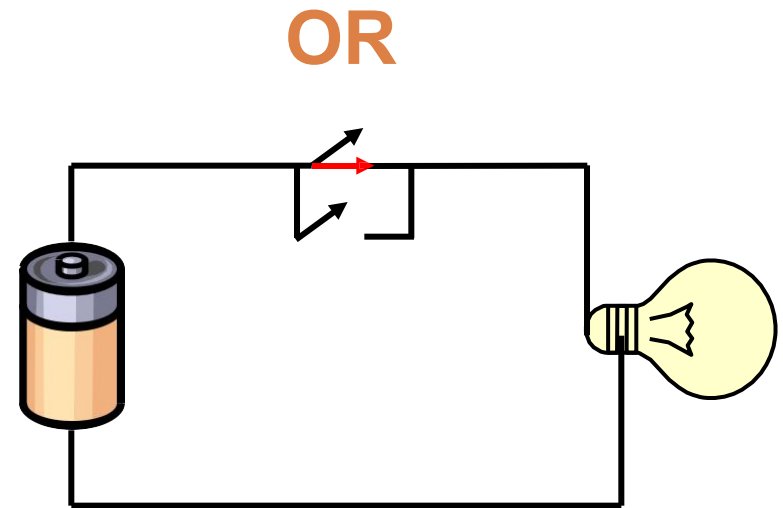
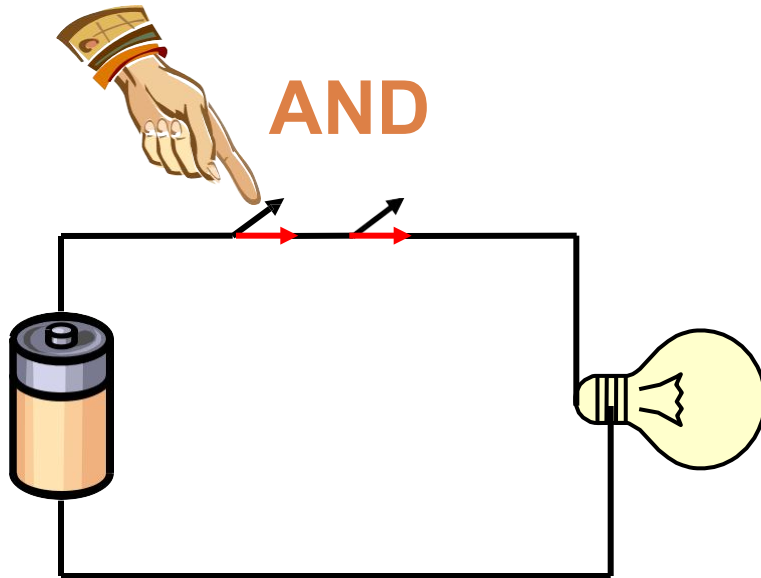
x	z
0	1
1	0

$$z = \bar{x} = x'$$



Switching Circuits

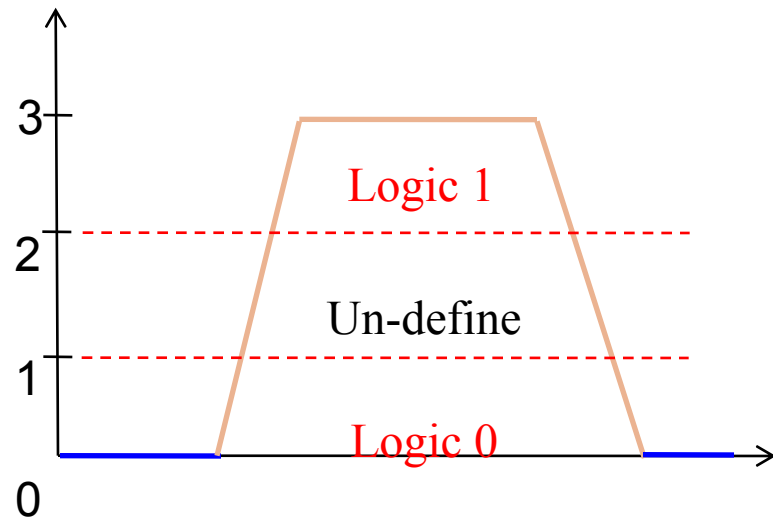
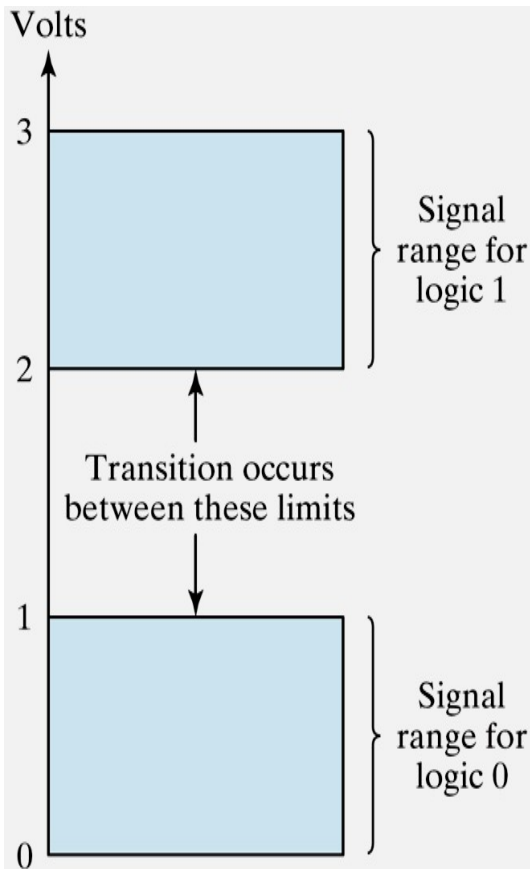
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Binary Logic

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- Logic gates
 - Example of binary signals



Example of binary signals

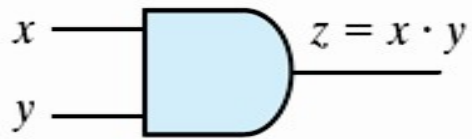


Binary Logic

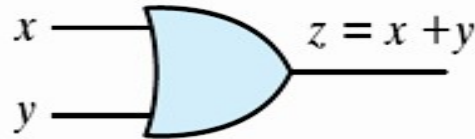
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□ Logic gates

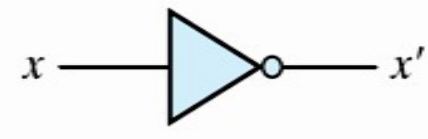
□ Graphic Symbols and Input-Output Signals for Logic gates:



(a) Two-input AND gate

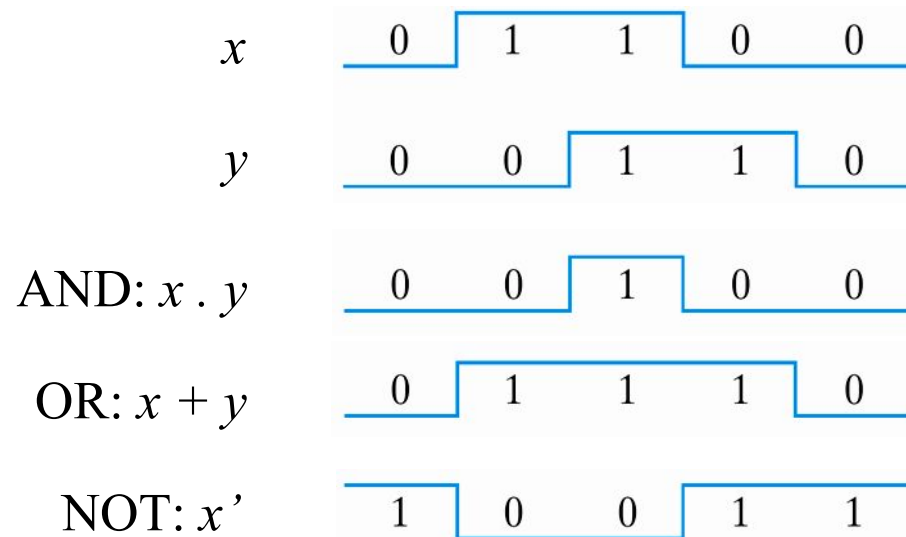


(b) Two-input OR gate



(c) NOT gate or inverter

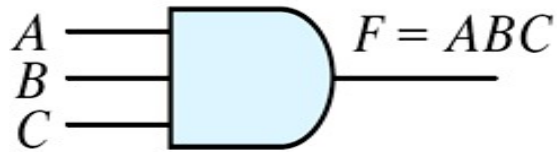
□ Example:



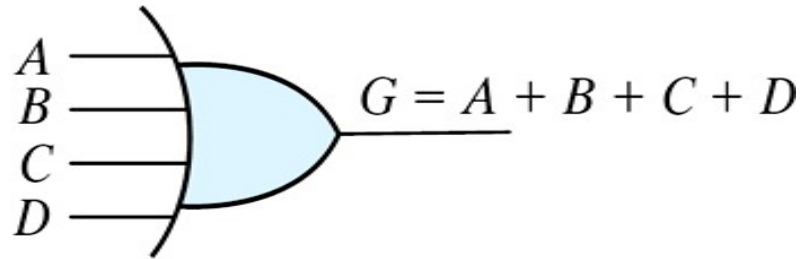
Binary Logic

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- Logic gates
 - Graphic Symbols and Input-Output Signals for Logic gates:



(a) Three-input AND gate



(b) Four-input OR gate

Gates with multiple inputs